

FloodProtect+

A parametric flood-recovery rider delivering liquidity for immediate expenses



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Source: Insurance Catastrophe Resilience Report 2024-25, Insurance Council of Australia
Image: Floods in Kimberley Region, Western Australia, January 2012
Department of Fire and Emergency Services WA

Australia's flood problem

Flood is the dominant natural peril in Australia

Flood damage accounts for more than **54%** of economic losses to Australian homes in recent years

Why Australia is uniquely exposed

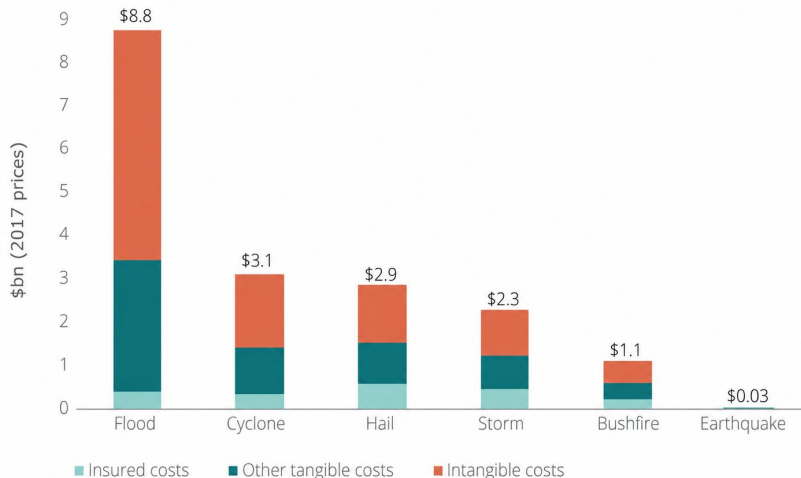
Geography: **90%** of Australia's population live on a coastal strip

Concentrated risk: Dense settlements are susceptible to accumulation risk, increasing premiums.

Multiple triggers: Tropical cyclones, flat inland topography, storm surges and monsoon seasons compound flood frequency

Flooding in Australia, 2020, IAG

Chart 2.2: Average annual economic cost, tangible costs and insured costs by disaster type*, 2007 to



*Disaster types are classified according to ICA definitions. See Appendix A.
Source: ICA (2017), Deloitte Access Economics estimates (2017)

The Problem

Despite its significance, flood remains one of Australia's least-insured perils.

Traditional coverage gap

Despite every major insurer offering flood cover, many households remain unprotected from flood risk.

The cost to cover flood events in high-risk areas push customers out of adequate protection

For a \$500,000 home in a high-risk zone, flood premiums alone can exceed **\$45,000 annually**.

The projected cost rise to flood estimated to be **+56%** by 2050 under moderate levels of climate change.

77% of 242,000 Australian properties at severe to extreme flood risk lack flood cover

Even with partial/full flood cover, traditional indemnity insurance cannot deliver

Under the General Insurance Code of Practice, insurers have up to **4 months** to decide on a claim

After the 2022 NSW and QLD floods, **35,000** claims remained unresolved for one year after the event.

Over **14,000** residents needed emergency accommodation

What Policyholders Actually Need



Affordable & accessible for homes in high flood-risk zones



Clear support benefits for indirect and incidental costs



Fast cash payout to bridge the support gap in days, not months

Parametric insurance

Addresses accessibility and claims delays by replacing loss assessment with objective triggers.

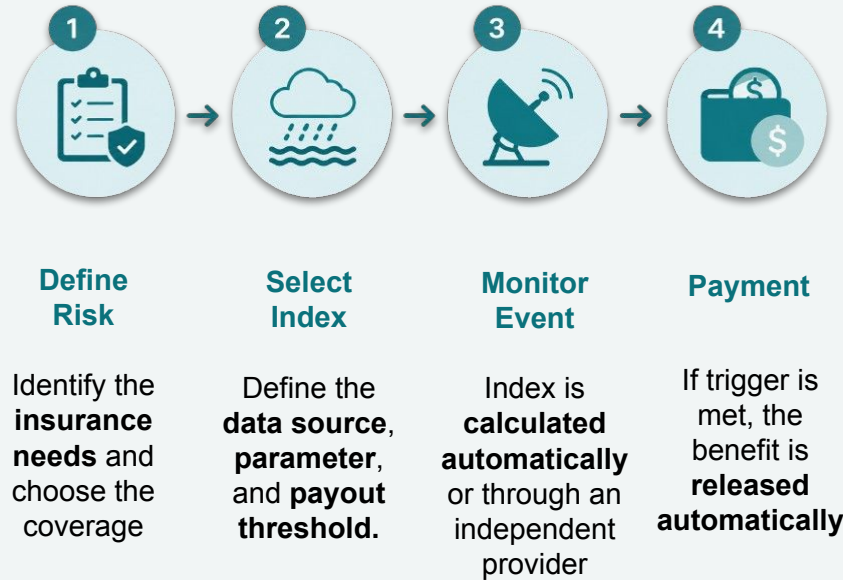
What is Parametric Insurance?

Parametric insurance pays a **predetermined amount** when an a defined parameter reaches a specified threshold

Unlike Traditional Insurance

- No loss assessment
- Predictable benefit
- Spending flexibility
- Automatic payout

How Does Parametric Insurance Work?



Properties of a Good Trigger



Objective

Not controlled by insurer or insured



Transparent

All parties can verify the index calculation



Timely

Measured quickly to support recovery



Relevant

Correlated with the insured's financial impact

FloodProtect+ product design

Fast-paying cover to supplement the secondary costs of flood

FloodProtect+ Rider

Predetermined benefit and trigger | Fast payment



Traditional Indemnity Base

Home and contents | Assessed loss | Higher limit

Purpose

Provide **immediate cash** after a flood event and minimise out-of-pocket costs.

Target

Homeowners in **flood-prone areas** with or without traditional flood cover

Trigger

Rainfall (and potentially other indices) exceed **calibrated threshold**

Benefit

Tiered fixed benefit comparable to disruption costs



Tier 1 - Minor - \$1,000

Cleanup and Sanitation Costs



Tier 2 - Moderate - \$3,000

7-day accommodation and incidentals



Tier 3 - Major - \$30,000

12 weeks housing and necessities

Flood event classification data: The Bureau of Meteorology, Insurance Council of Australia
Benefit cost benchmarks: Australian Bureau of Statistics, Local contractor costs

Evidence-based trigger calibration

Pilot case study: Lismore / Northern Rivers (NSW) and Ipswich (QLD) station zones

Calibration matrix

	No flood	Minor flood	Moderate flood	Major flood
No payout	Aligned		Underpay / miss	
Tier 1 payout		Aligned		
Tier 2 payout	Overpay / false payout		Aligned	
Tier 3 payout				Aligned

Diagonal = aligned | Darker shade = greater mismatch (basis risk)

Sources: BoM station rainfall and river gauges;
official flood-event records in Ipswich, QLD and Lismore, NSW

Calibration Steps

Build Rainfall Indices

1-day, rolling **3-day** and **7-day** rainfall indices

Define Episodes and Labels

Wet periods are clustered into **episodes**

Calibrate Tier Thresholds

Align trigger frequency with **true flood events**

Payment Rule

The **highest tier** per rainfall episode is paid

Calibration Criteria

- Minimise missed true flood events
- Higher weights assigned to under-payment

Selected

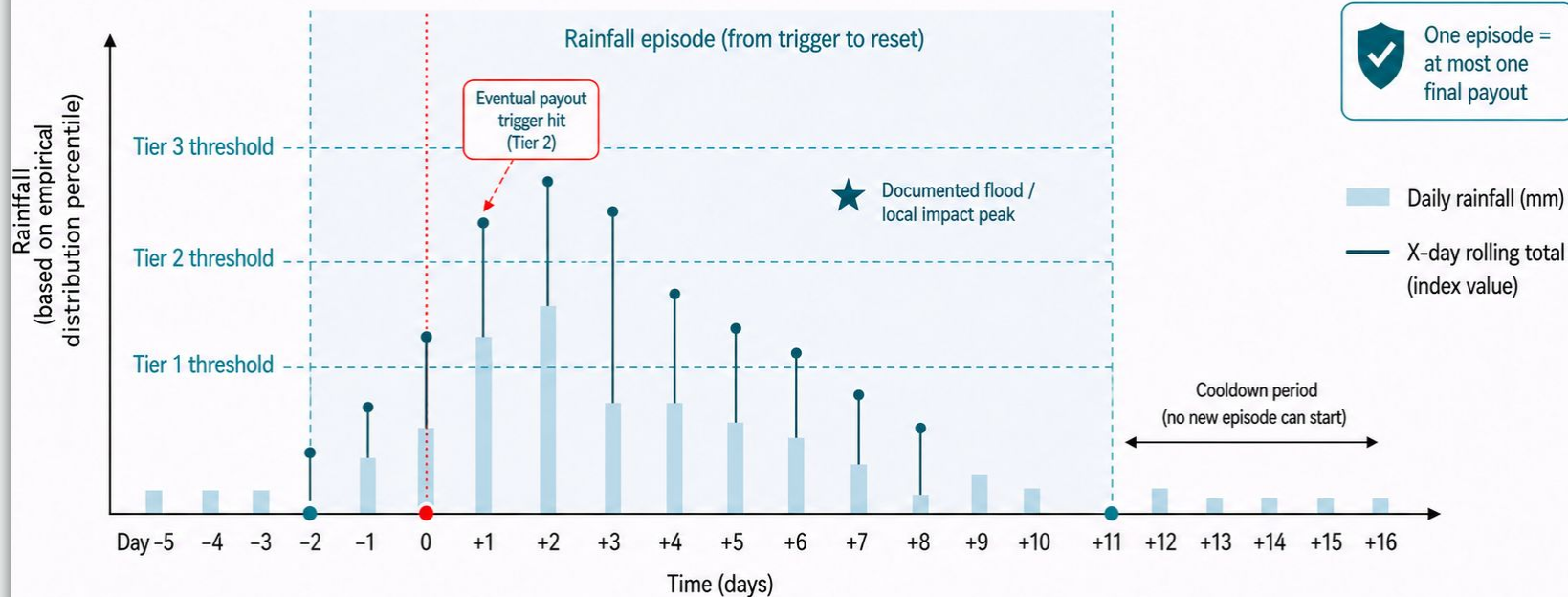
- Lismore 7-day window | Ipswich 3-day window
- Rainfall percentile thresholds: 99.2 / 99.5 / 99.85
- Cooldown rule: 21 days / 42 days / 90 days

Illustrative rainfall episode, trigger and tiered parametric payout

1 Payout development once the trigger condition is first met

2 Final payout tier = highest threshold reached during the episode

3 Episode ends after the reset period



Simulated payout experience and premium setting

Pricing and affordability simulation: repeatedly sampled rainfall paths over 30 policy years

Premium Composition



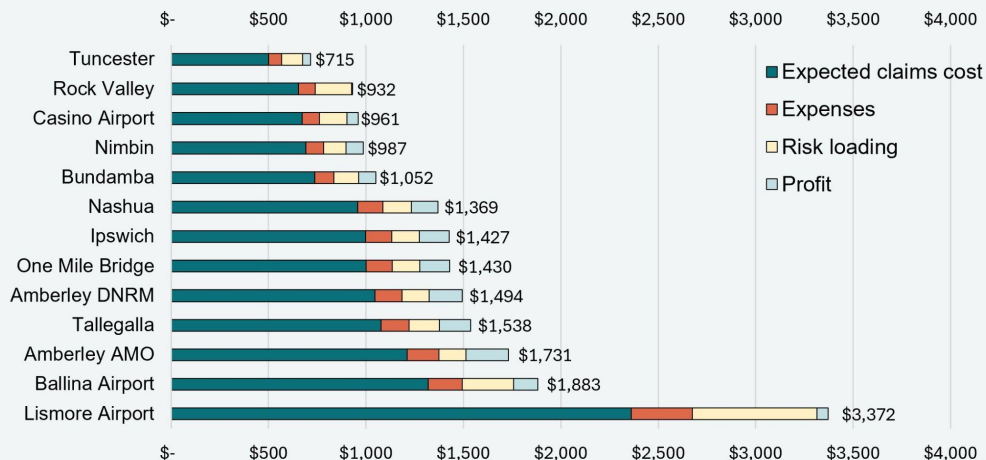
Expected claims cost

Expenses

Risk Loading

Profit

Station-zone Premium Breakdown



Remarks:

The case study was conducted on selected high-risk study areas and represent a hypothetical MVP, rather than a fully diversified commercial portfolio or market-wide pricing outcome.

Premium methodology

- Pricing target loss ratio of **70%**
- Expense ratio of **10%**
 - typical for parametric products
 - significantly lower than traditional indemnity (**25%-35%**)
- Risk loading approx. **13%**
 - calculated using the cost-of-capital approach
 - Required additional capital (**95% VaR less expected loss**) at **6% cost of capital**.
- Profit margin approx. **7%**, after accounting for expenses and risk loading.

Highlights:

High payout probability: Ballina Airport (64.6%)

- Annual probabilities: **Tier 1 50.4%**, Tier 2 12.7%, Tier 3 0.7%

Low payout probability: Tuncester (18.6%)

- Annual probabilities: Tier 1 3.0%, **Tier 2 14.5%**, Tier 3 <0.1%

Caution: Lismore Airport

- Unusually **frequent Tier 3 payouts** driven by near-threshold historical extremes in the data sampling process.
- Exposed to extreme or prolonged rainfall risk.
- Estimated premium for full indemnity cover ranges **\$15,000 - \$25,000 p.a.** (based on recent parliamentary inquiry)

Note: payout probability is the probability of at least one paid episode in a simulated policy-year

Stress testing

Exposure-weighted aggregate outcomes (15,900 policies across 12 station-zones).

Scenario 1

Severe cross-region event

One scaled 21-day severe rainfall block is injected into both pilot regions in simulation, then the full payout simulation is rerun.

Scenario 2

Misspecified, Repeated Event Frequency

Two or three (more frequent) severe blocks are injected randomly, with episode cooldowns and annual payout caps active.

Scenario 3

Upside and downside basis risks

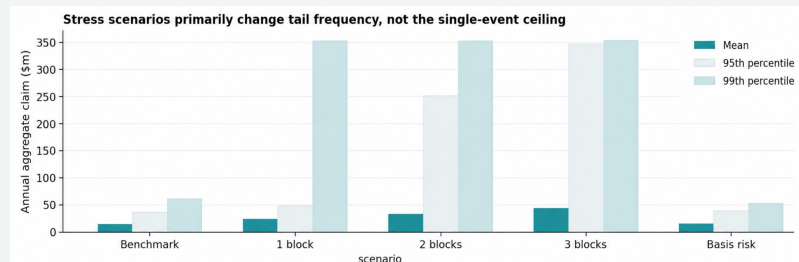
Impact /no-impact labels are re-assigned and attached to different wet periods to estimate false-positive claim leakage and false-negative support gap under stress.

IMPLEMENTATION PRINCIPLE

The scenario-based testing stress changes the rainfall pathways or event impact labels; the **same rolling-index trigger, reset/cooldown period, annual payout caps and station-zone aggregation** are rerun unchanged.

Portfolio summary

Mean, 95th and 99th percentile annual aggregate claims.



- One severe cross-region event block clearly transforms the tails: (99th percentile) **\$61.2m → \$352.6m**.
- More frequent, unprecedented events stressed payout into the 95th / 90th percentiles.
- Result under basis-risk re-simulation is similar to benchmark

Aggregate claim comparison

Simulated annual aggregate claim distribution, PML and regional contribution of mean change.

Scenario	Mean	95%	99%	PML	Regional contribution to change
Benchmark	\$14.41m (ref.)	\$36.70m	\$61.20m	\$153.30m	Composition: Northern Rivers / Lismore 42.3%; Ipswich 57.7%
1 severe block	\$24.35m (+68.9%)	\$49.21m	\$352.60m	\$356.60m	Increase: Northern Rivers / Lismore 50.4%; Ipswich 49.6%
2 severe blocks	\$33.41m (+131.8%)	\$252.20m	\$353.00m	\$358.80m	Increase: Northern Rivers / Lismore 50.5%; Ipswich 49.5%
3 severe blocks	\$43.63m (+202.7%)	\$347.40m	\$354.20m	\$357.60m	Increase: Northern Rivers / Lismore 51.1%; Ipswich 48.9%
Basis-risk resim.	\$14.81m (+2.8%)	\$38.80m	\$53.20m	\$176.60m	Northern Rivers / Lismore 40.8%; Ipswich 59.2%

Assumptions & limitations

Key assumptions and limitations across the modelling chain: data → trigger → product → portfolio

1 Data & study design

Assumptions

- Sources: BoM rainfall and river gauges, supported by official flood records
- Granularity: station and gauge data represent local flood exposure
- Pilot areas: Lismore (NSW) and Ipswich (QLD)
- Event labels: historical flood records and gauge observations

Data Risk

- **Sparse or incomplete** event data
- Station-level data may **miss property-level variation**
- **Limited transferability** to lower-risk areas

2 Trigger Model

Assumptions

- Index: rolling rainfall totals proxy flood impact
- Windows: 3-day and 7-day accumulation by pilot area
- Thresholds: empirical percentiles from historical rainfall
- Return periods: estimated from approximately 20 years of data

Basis Risk

- Rainfall alone cannot represent all flood types
- Fixed windows may **miss event timing**
- Historical data may **understate** climate trends and extremes

3 Product Design

Assumptions

- Payout: three tiers reflect event severity and recovery needs
- Cost basis: calibrated to average immediate recovery costs
- Pilot scope: selected flood-prone regions
- Controls: payout **caps** and **cooldown periods** limit repeated claims

Coverage Risk

- Avg costs **miss** household and property **differences**
- Higher cover must be balanced against affordability
- Add-on design for well-protected homes **requires testing**

4 Portfolio & Pricing

Assumptions

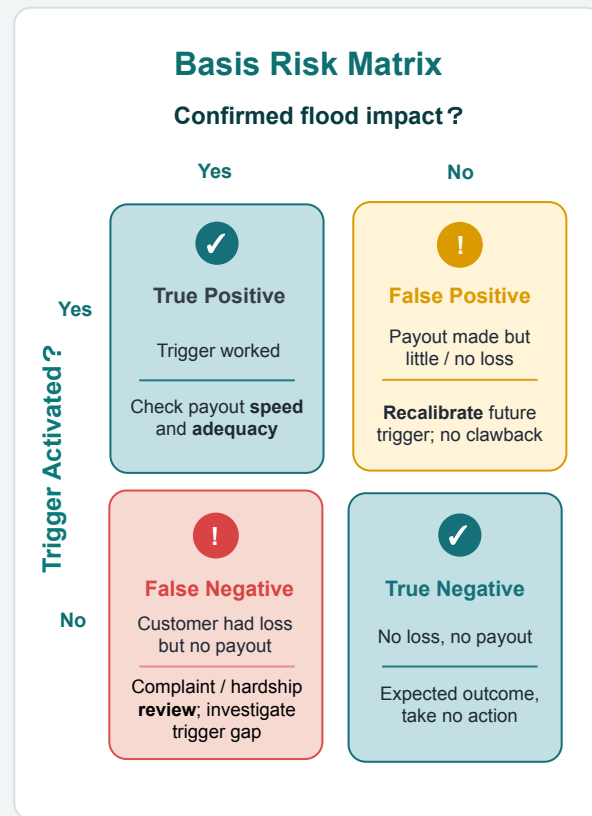
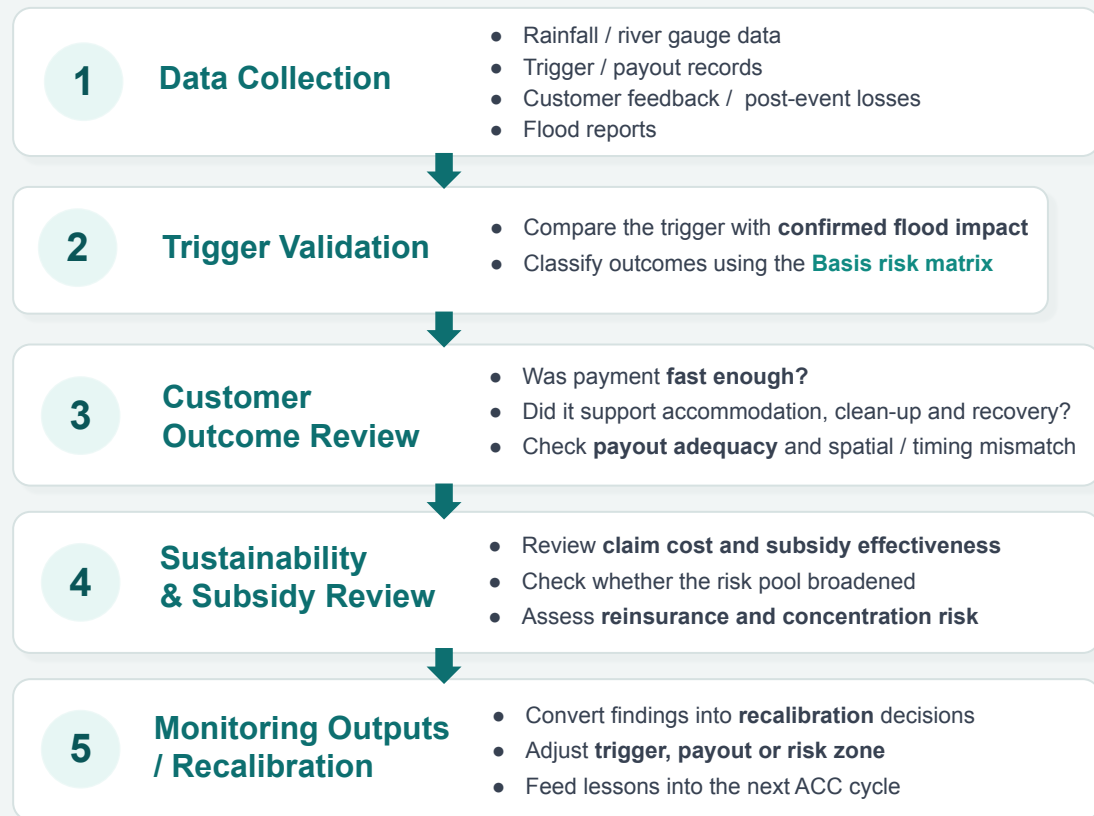
- Simulation: block bootstrap preserves weather dependence
- Exposure: residential density proxies policy concentration
- Diversification: assumes sales across 10+ zones in two states

Concentration Risk

- Density misses property value, resilience and insurance uptake
- Correlated flood systems may **overstate diversification**
- Sales mix is **concentrated** in mid- and high-risk areas

Monitoring process overview

How we validate the flood pilot and translate monitoring signals into product improvement



Monitoring outputs / recalibration decisions

How monitoring signals feed into the next Actuarial Control Cycle and prepare the pilot for multi-peril expansion

Monitoring signal

Recalibration response

Repeated False Negatives



Hardship review; **investigate trigger gap**; add station/gauge or extend rainfall window.

Repeated False Positives



Recalibrate future trigger or refine risk zone; no clawback.

Payout Mismatch



Adjust payout tiers or offer a higher-cover option.

Spatial / Timing Mismatch



Use closer data, smaller risk zones or a different rainfall window.

Concentration or Subsidy Pressure



Review pricing, reinsurance, **sales mix** and subsidy allocation.

Data Reliability Issue



Add **backup data**, missing-data rules and an auditable payout process.

ACC feedback loop Experience → assumptions → trigger design → pricing / subsidy → monitoring

Future peril expansion readiness

Test whether the flood platform can scale to other perils.

Data integration



Can storm, cyclone or bushfire data be **integrated**?

Overlap rules



How are events triggering multiple perils treated?

Accumulation risk



Are flood and storm triggers highly **correlated**?

Customer wording



Is flood-only coverage **clear** to customers?

Subsidy & Reinsurance



Is support **sustainable** under multi-peril cover?

Flood pilot → multi-peril policy design